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Effect of micro-aeration and inoculum type on the biodegradation of lignocellulosic substrate

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HIGHLIGHTS

- Micro-aerobic conditions were applied to enhance the methane yield of wheat straw.
- Oxygen load, pulse frequency and micro-aeration period were the examined variables.
- Extended hydrolysis and solubilization of straw were achieved due to micro-aeration.
- The most efficient micro-aeration strategy increased the methane yield by 7.2%.
- AD process was susceptible to instability at oxygen loads higher than 10 mL O_2/g VS.

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ABSTRACT

The effect of various micro-aeration strategies on the anaerobic digestion (AD) of wheat straw was thoroughly examined using a mixture of inocula, containing compost and well digested sludge from biogas plant. The aim was to determine the most efficient oxygen load, pulse repetition and treatment duration, resulting in the highest methane production. The oxygen load had the largest impact on the biodegradability of straw, among the examined variables. More specifically, a micro-aeration intensity of 10 mL O_2/g VS was identified as the critical threshold above which the AD performance was more susceptible to instability. The highest enhancement in biogas production was achieved by injecting 5 mL O_2/g VS for a consecutive 3-day treatment period, presenting a 7.2% increase compared to the untreated wheat straw. Nevertheless, the results from optimisation case study indicated a higher increase of 9% by injecting 7.3 mL O_2/g VS, distributed in 2 pulses during a slightly shorter treatment period (i.e. 47 h).

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1. Introduction

During the recent years, the annual production of wheat is steadily increasing, reaching 729 Mt in 2014 (Faostat, 2016) with 20% of the global amount being produced in Europe. Unfortunately, the entire harvest cannot be utilized for food production, resulting in a residue to crop ratio of 1.3:1.0 (Milbrandt, 2005). Thus, the significant amount of biomass which is typically disposed could be exploited for alternative purposes, such as energy generation in the form of biogas via the anaerobic digestion (AD) process.

It is well known that pretreatment methods are typically applied to all lignocellulosic substrates in order achieve efficient AD performance. These methods target to improve the hydrolysis step and thus enhance substrate's degradability (Carrere et al., 2015). However, the existing pretreatments are usually associated with specific drawbacks, such as the need for an increased energy supply (e.g. for mechanical pretreatment), the possibility of inhibitor formation (e.g. for chemical pretreatment) or difficulties in the application at an industrial scale (e.g. for biological pretreatment) (Zheng et al., 2014). As a result, even if these methods can initially increase the biogas production, the final economic balance is not always positive. Alternative techniques are lately developed to enhance the hydrolysis of complex organic polymers, without the need for increased energy carriers or without the production of undesirable by-products (e.g. furfural, 5-hydroxymethylfurfural and phenols).

Accordingly, providing the digester with small amounts of oxygen or air has been considered an alternative tool to improve the overall AD (Krayzelova et al., 2015). The synthesis of unique extracellular hydrolytic enzymes has been reported to be higher in the presence of oxygen (Lim and Wang, 2013); thus, the creation of micro-aerobic conditions is an interesting solution to promote

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the activity of aerobic and facultative microorganisms which can enhance solubilisation of the organic matter (Botheju et al., 2010).

Since the efficiency of micro-aeration is mainly based on the existence of specific hydrolytic microbes, it is important to ensure that these members are present in the inoculum in order to secrete the needed extracellular enzymes. Thus, in a semi-aerobic environment, as for instance in the one created during the composting process, the desired bacterial consortium could be found in abundance (Beffa et al., 1996). Indeed, the existence and activity of this microbial group are mandatory, in order to achieve a considerable disintegration of lignocellulosic wastes at composting facilities (Jurado et al., 2015). From this point of view, the utilization of an active inoculum from an AD process, which contains all the necessary members of the microbial community (i.e. strictly anaerobic) for organic matter degradation, combined with an inoculum derived from the hot phase of composting, which includes a more specialized microflora (i.e. facultative anaerobic) could be a solution for an enhanced AD process. As concluded by Scherer and Neumann (2013), the methanation process of lignocellulosic sugar beet silage was considerably improved when the reactors were inoculated with a mixture of methanogenic and compost inocula. This combination of inocula would result in a balanced microbial consortium for the decomposition of recalcitrant organic matter with cooperative interspecies activity and would finally result in improved process efficiency.

Therefore, the enrichment of AD microbiome with an aerobic inoculum, along with the application of micro-aeration could lead to improved biodegradation of complex substrates (e.g. extended hydrolysis). However, the effect of micro-aeration on AD remains controversial, majorly due to the adverse impacts that appear in cases that oxygen is not supplied properly. More specifically, it is crucial to control the oxygen concentration at suitable levels, so as to avoid increased oxygen respiration, which will be combined with a decreased availability of substrate for the rest members of the complex community (Zhu et al., 2009). At the same time, increased substrate solubilization should be achieved without provoking inhibition of the strictly anaerobic methanogenic Archaea which perform methanogenesis and are extremely sensitive to oxygen (Botheju and Bakke, 2011). Nevertheless, research on sludge-based reactors proved that there is a spatial distribution of Archaea, with the latter residing mainly at a distance from oxygen exposure; in contrast to facultative microbes which can tolerate limited amounts of oxygen (Krayzelova et al., 2015). Thus, a proper micro-aeration strategy should not increase the toxicity to methanogenic Archaea, as oxygen can be efficiently utilized by facultative microbes to favour the activity (Jenicek et al., 2008; Tartakovsky et al., 2011).

A number of different oxygenation intensities and injection techniques have already been examined for improved disintegration of various cellulosic substrates (Botheju et al., 2010; Diaz et al., 2011; Jagadabhi et al., 2010; Lim and Wang, 2013; Zhu et al., 2009). Nevertheless, the conclusions of the different studies are contradictory and no clear evidence exists, regarding the appropriate strategy of oxygen loading that would result in maximized hydrolysis and subsequently, increased anaerobic degradability. For example, even though there is a number of studies which have found positive impacts of micro-aeration (Botheju et al., 2010; Lim and Wang, 2013; Tartakovsky et al., 2011), Johansen and Bakke (2006) found that the supply of a digester with oxygen deteriorated the overall AD process, leading to reduced methane yield. In fact, the previous investigations were performed at different micro-aeration intensities, techniques or loads. Thus, a systematic analysis of different micro-aeration strategies would identify the oxygen load thresholds and thus, improve the knowledge about the effect of micro-aeration on the AD process with respect to increased biodegradation of wheat straw.

Previous studies have evaluated the effect of micro-aeration on the separate steps of the AD process. However, no consistent conclusion is yet available. The present study aims to elucidate the effect of the micro-aerobic conditions on wheat straw solubilization and biodegradability. Specifically, various oxygen loads, pulse frequencies and micro-aeration periods in different combinations were investigated. Moreover, the effect of inoculum source on substrate's biodegradability was elucidated, using a mixture of methanogenic and compost inoculum.

2. Materials and methods

2.1. Inocula

Effluent from Snertinge biogas plant (Zealand, Denmark) operating at thermophilic conditions was used as AD inoculum. After its arrival at the laboratory, the collected inoculum was sieved to remove the particles and, subsequently, was stored in a thermophilic incubator for one week, in order to reduce its biogas production.

The compost was obtained from a composting plant in Kongens Lyngby (Zealand, Denmark), where organic materials such as plants and garden wastes are used as feedstock. The sample was derived from the thermophilic phase of the process and was between 5 and 6 weeks old. Similarly to the AD inoculum, the compost inoculum was sieved to discard large particles, prior to usage. Subsequently, a suspension of similar VS content as the AD inoculum was prepared with anoxic tap water. The characteristics of both inocula are presented in Table 1.

2.2. Wheat straw

Wheat straw was grown and harvested from a field in the surrounding area of Copenhagen (Zealand, Denmark). Biomass was cut in length less than 0.5 cm, using a cutting mill (SM 200, Retsch GmbH, Germany). For chemical characterization, the biomass was further grounded to 2 mm. The total solids (TS) and volatile solids (VS) of the lignocellulosic biomass were 928.39 ± 0.01 g/kg and 867.22 ± 3.85 g/kg, respectively. The biomass composed of 42.02 ± 0.72 %TS, 30.75 ± 0.55 %TS, 26.73 ± 2.35 %TS and 1.96 ± 0.40 %TS of cellulose, hemicellulose, lignin and proteins, respectively. The chemical oxygen demand (COD), Total Kjeldahl Nitrogen (TKN) and C/N ratio were 1060.92 ± 43.26 g/kg TS, 3.90 ± 0.07 g/kg and 103, respectively.

2.3. Experimental design and statistical analysis

Using response surface methodology, the interactions among different variables can reveal the optimum pretreatment conditions by conducting the minimum number of experimental combi-

Table 1
Main characteristics of AD and compost inoculum.

Characteristics	AD inoculum	Compost inoculum
pH	8.21	8.20
TS (g/L)	27.47 ± 0.24	27.05 ± 0.68
VS (g/L)	17.11 ± 0.12	17.01 ± 0.32
TKN (g/L)	3.63 ± 0.13	0.39 ± 0.07
NH_4^+-N (g/L)	3.51 ± 0.12	0.07 ± 0.00
Total volatile fatty acids (VFA) (mg/L)	93.40 ± 13.08	43.80 ± 3.25
Acetate (mg/L)	70.71 ± 10.96	24.56 ± 2.30
Propionate (mg/L)	10.55 ± 0.94	5.94 ± 0.36
Isobutyrate (mg/L)	2.10 ± 0.10	0.00 ± 0.00
Butyrate (mg/L)	4.17 ± 0.83	0.00 ± 0.00
Isovalerate (mg/L)	2.08 ± 0.24	3.41 ± 0.58
Hexanoate (mg/L)	1.77 ± 0.64	9.89 ± 0.45

nations (NIST/SEMATECH, 2012). Central Composite Design–Face Centered (CCF) was used to define the optimum micro-aeration strategy in order to enhance the biodegradability of wheat straw. In the present model, the factorial design levels were coded from −1 to +1 and all points had a distance of ±1 of the factorial central point. Oxygen load (mL O_2/g VS), pulse (injections' repeatability) and micro-aeration period (days) were the examined variables at 3 different levels (Table 2). The examined overall O_2 loads were of 5, 10, 15 mL O_2/g VS, the O_2 volume was shared in 1, 2, 3 injections and additionally, the micro-aeration period lasted for 1, 2, 3 days. The O_2 loads were carefully selected in order to be in the range of previous works performed on different substrates (Fu et al., 2015; Lim and Wang, 2013). Coded and real values for the examined variables are shown in Table 2. The results were analyzed using analysis of variance (ANOVA) by Design-Expert software. The functional relationships between methane yield and the examined variables were described by formulating the coefficients of a second-order polynomial model, based on the Biochemical Methane Potential (BMP) results. The fit of the model was evaluated by the coefficient of determination (R^2), adjusted R^2, adequate precision and 'lack of fit' F-tests. Moreover, the effect of micro-aeration strategies on pH, oxidation–reduction potential (ORP), VFA, soluble chemical oxygen demand (sCOD) and biomethanation was evaluated using Graphpad Prism 5 software package (Graphpad Software, Inc., San Diego, CA). Descriptive statistics were conducted for all data and mean values and standard deviations were calculated. The comparison of quantitative variables among the examined treatments was performed by one-way ANOVA, followed by Tukey post hoc test ($p < 0.05$).

2.4. Experimental procedure

Batch assays were conducted in thermophilic conditions (55 °C) in triplicates according to the guidelines of Angelidaki et al. (2009). The used glass reactors had a total volume of 547 mL and working volume of 200 mL. Initially, a BMP set digesting 1 g wheat straw using only 160 mL AD inoculum and a BMP set using the mixture of inocula (i.e. 80 mL AD and 80 mL compost) were elucidated without applying micro-aeration. In both BMP sets, 40 mL water were also added and thus, the organic load was calculated to be 4.33 g VS/L. Subsequently, the micro-aeration treatments were conducted as described in the previous subchapter. For each set, blank reactors containing 80 mL AD inoculum, 80 mL compost inoculum and 40 mL water (without addition of wheat straw) with the same micro-aeration strategy were conducted in order to examine possible effects of micro-aeration on the inoculum mixture.

The micro-aerated reactors were supplied with 1 g wheat straw, 80 mL AD inoculum, 80 mL compost inoculum and 40 mL water. Thus, the organic load of the batch assays was equal to the non-aerated reactors. Before incubation, the reactors were flushed with N_2 gas to ensure anaerobic conditions and were closed using rubber stoppers, sealed with aluminum crimps. Finally, oxygen was added in the reactors according to the micro-aeration strategies. Pure oxygen was added directly into the liquid phase of the reactor. By addition of oxygen, oxidation of sulphide, which is toxic to archaeal community, occurs as an extra benefit of the micro-aeration (Krayzelova et al., 2015). Methane concentration was monitored twice per week at the head space of the reactors, until cease of production was observed. In total, the digestion period lasted 35 days and the final methane yields were calculated using the average values among replicate reactors. The methane yield of batch assays was always calculated after subtraction of background biomethanation as measured from the corresponding blank reactors. Moreover, at the end of each micro-aeration period, hydrolysate/digestate samples were taken from each batch set, in order to evaluate the effect of different O_2 loads at pH, VFAs and sCOD. Untreated samples were collected for the same time intervals as a baseline comparison. The physicochemical parameters (i.e. pH, VFA and sCOD) were also measured at the end of the digestion period.

Table 2
Variables used in the CCF experimental design and design levels.

Variables	Level −1	Level 0	Level +1
Oxygen load, mL O_2/g VS	5	10	15
Pulse, injections' repeatability	1	2	3
Micro-aeration period, days	1	2	3

2.5. Analytical methods

The TS, VS, TKN, NH_4^+-N and COD were determined according to the Standard Methods (APHA, 2005). For the determination of soluble COD (sCOD), the supernatant of the centrifuged samples at 12,000 rpm for 10 min was used. The pH and ORP variations were determined with a benchtop pH/ORP meter (FEP20, Mettler Toledo). Elemental analysis of wheat straw was conducted using a vario MACRO cube (CHNOS Elemental Analyzer, Hanau-Germany). Analysis of carbohydrates and lignin content was conducted according to the guidelines of the National Renewable Energy Laboratory (Sluiter et al., 2011). Ion Chromatography (Dionex ICS-5000, Thermo Scientific) was used to determine carbohydrates content. Separation of sugars was carried out by a CarboPak PA1 analytical column (250 × 4 mm) using 100% Milli q water (eluent A), 1 M NaOH (eluent B) and 100 mM NaOH (eluent C). The mobile phase was set at a constant flow rate of 1.0 mL/min. The methane concentration in batch reactors was measured using a gas chromatograph (Shimadzu GC-8A, Tokyo-Japan) with a glass column (2 m, 5 mm OD, 2.6 mm ID) packed with Porapak Q 80/100 mesh (Supelco, Bellefonte, PA, USA) and with a flame ionization detector (FID) as previously described by Kougias et al. (2013). The VFA composition was determined by a gas chromatograph (Shimadzu GC-2010, Kyoto, Japan) with a FID as described by Kougias et al. (2015).

3. Results and discussion

3.1. Effect of inoculum type on the AD of wheat straw

Initially, the effect of compost as inoculum was elucidated on wheat straw anaerobic biodegradability (without addition of oxygen). The results showed that the practical methane yield of untreated wheat straw using AD inoculum (WS) (255 ± 17 mL CH_4/g VS) did not differ significantly ($p > 0.05$) from the corresponding yield using the mixture of inocula (0–0–0) (257 ± 6 mL CH_4/g VS) (Fig. 1). Moreover, the methane content in the biogas for both BMP sets did not differ during incubation period (65–67%). Hence, the usage of composting inoculum did not lead to any synergistic or antagonistic effect in the final methane yield. These results are contradicting previous findings obtained by Scherer and Neumann (2013) who concluded that the methane production of sugar beet was increased by 27% by supplementing anaerobic inoculum (i.e. digested sewage sludge) with aerobic inoculum (i.e. plant litter-compost). Additionally, in the present study neither the methane production rate nor the duration of AD were favoured.

Hence, it was concluded that no impact was observed on methanation due to the enrichment of the microbiome. However, the micro-aeration experiments were conducted using the mixture

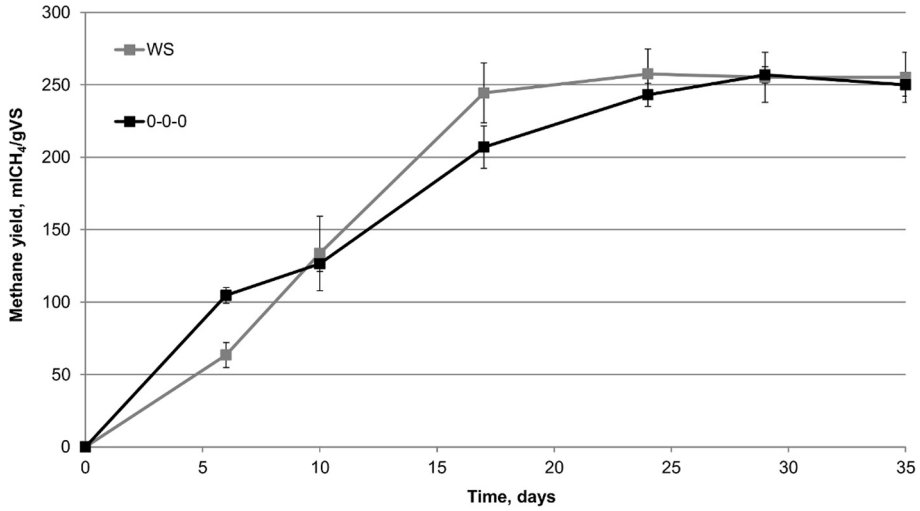


Fig. 1. Accumulated methane yield of batch reactors digesting wheat straw using only AD inoculum (i.e. WS) and a mixture of inocula (i.e. 0–0–0).

of inocula, in order to evaluate whether the action of aerobic and facultative microorganisms and fungi could have been favoured due to the presence of oxygen. Subsequently, the overall biodegradation could be enhanced without provoking process inhibition, as oxytolerant Archaea are typically found in similar AD systems (Fu et al., 2016).

3.2. Effect of micro-aeration on ORP, pH, sCOD and VFA using a mixture of inocula

The impact of micro-aeration on the physicochemical characteristics of BMP assays was monitored during the distinct period in which the oxygen was injected, in order to reveal potential differences among the treated and untreated samples. Therefore, the effects of the diverse micro-aeration strategies on pH, sCOD and VFA during the oxygen addition period are presented in Fig. 2a–c, respectively. Additionally, all the aforementioned parameters were measured at the end of the AD period (i.e. 35th day). The influence of oxygen addition into the biogas reactors was initially evaluated by measuring the ORP values during the treatment period. Regarding the micro-aerated reactors, the values ranged from −65 to −72 mV. Similarly, the non-aerated reactors had ORP values of −63 to −82 mV. These measurements indicated that all batches had anaerobic zones and hence, the micro-aeration did not create an oxidizing environment inside the reactors (Lim and Wang, 2013). Subsequently, at the end of incubation period, the ORPs varied from −250 to −300 mV; these values are within the normal range for strictly anaerobic conditions (Jenicek et al., 2008).

As depicted in Fig. 2a, the pH of micro-aerated samples decreased more rapidly compared to their corresponding untreated samples (i.e. black columns) during the initial 2-day treatment period. On the contrary, no distinct differences were visible on the third day, which was the last day of the micro-aeration period (i.e. 5–3–3, 10–2–3 and 15–3–3), as the pH levels of both untreated and treated did not differ significantly ($p > 0.05$). The small differences can be attributed to the usage of AD inoculum which can provide the system with high buffer capacity in order to maintain the pH values relatively unchanged (Charles et al., 2009). Likewise, after 35 days of AD, in all the batch assays, the pH varied insignificantly ($p > 0.05$) (~7.46).

Regarding the solubilization of organic matter, clear changes were observed between non-aerated and micro-aerated samples (Fig. 2b). Generally, the treated samples had increased sCOD content compared to the untreated at the end of the micro-aeration period, indicating the impact of oxygen on lignocellulose deconstruction (Jagadabhi et al., 2010). Following this concept, Lim and Wang (2013) found that the sCOD of organic matter was enhanced by more than 50% due to the injection of 37.5 mL O_2/L_R during four consecutive days. The increased solubilization could be possibly attributed to the presence of fungi in the aerobic inoculum, which are capable of decomposing organic biomass under anaerobic conditions, after an initial step of sufficient oxygen exposure (Peces et al., 2016). In contrary, no distinct changes were detected among micro-aerated samples. At the end of the digestion period, the sCOD of micro-aerated batch assays showed no discernible differences (<1.5 g/L) (data not shown). Hence, the initially extended hydrolyzed organic matter has undergone efficiently the rest of the AD steps and was mineralized to CH_4 and CO_2. Comparatively, the sCOD values of untreated samples were slightly higher (~2 g/L) concluding that the biomass was disintegrated to a decreased extent compared to the micro-aerated samples.

Fig. 2c presents the concentrations of acetate and also, total VFAs produced, at the end of each micro-aeration strategy. The total VFAs represent the sum of acetate, propionate, isobutyrate, butyrate, iso-valerate, valerate and hexanoate concentrations. It was shown that the micro-aeration deeply influenced the VFA concentration leading to an accumulation of VFAs shortly after the injection of the oxygen pulse (i.e. recorded after the second day). As shown in Fig. 2c, acetate represented the highest portion of VFAs, which is subsequently used as substrate by the archaeal community. Conversely, more complex fatty acids were not accumulated. For example, propionic acid production is favored in less anaerobic environments with higher redox conditions (Sawatdeenarunat et al., 2015). Thus, it is implied that the examined oxygen load did not provoke an important shift towards unwanted higher redox potentials, which can be also concluded from the negligible changes in ORP values that presented above. Moreover, the proportions of acetate compared to total VFAs were higher in all cases in comparison with the values of untreated samples. The present results are in agreement with Lim and Wang (2013) who investigated the application of micro-aeration on a relatively easier feedstock to anaerobic biodegradability, compared to the examined substrate, and found that the injection of oxygen enhanced acetate formation and decreased propionate accumulation. Thus, it is validated that oxygen promoted an increased acetate concentration and enhanced the breakdown or oxidation of the less favorable VFA form for methanogenesis (i.e. propionate) in the hydrolysate. Additionally, the VFAs composition was measured

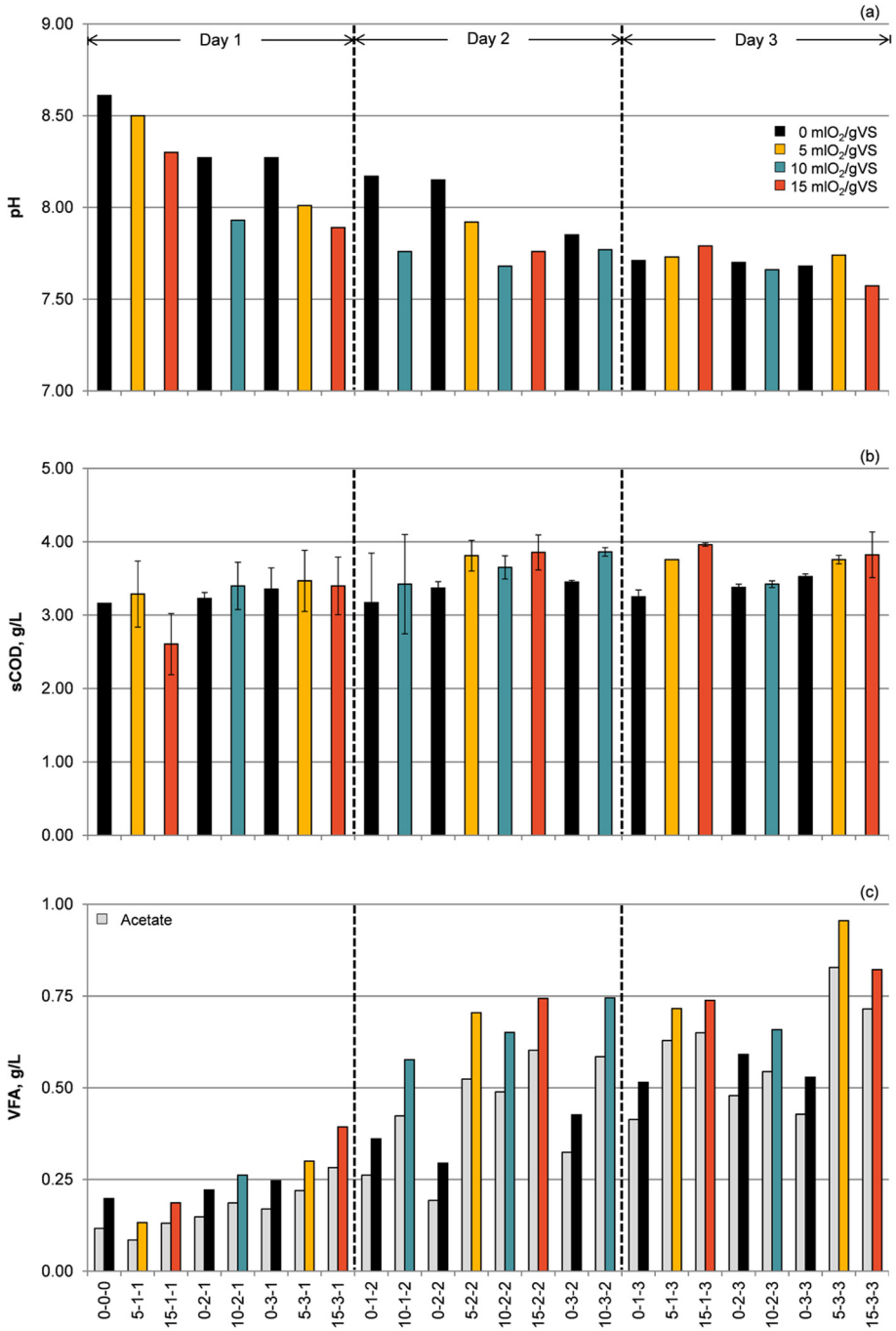


Fig. 2. Effect of micro-aeration on pH (a), sCOD (b) and VFA (c). The treatments are denoted with the amount of oxygen load (mL O_2/g VS) – number of pulses – micro-aeration period in days (e.g. 5–3–1). The black columns represent always the control operation in comparison to the corresponding micro-aerated samples during the same time interval.

again in the last day of the digestion period, revealing values lower than 40 mg/L for all samples (data not shown). Thus, it is concluded that the hydrolyzed organic matter was efficiently transformed to biogas and no inhibition to methanogens occurred.

3.3. Effect of micro-aeration on methane yield using a mixture of inocula

3.3.1. BMP experiments

The methane yields of untreated and micro-aerated wheat straw using a mixture of the two inocula are presented in Fig. 3. The results demonstrate that different micro-aeration strategies affect remarkably the methane potential of the lignocellulosic substrate.

The highest methane yield enhancement was achieved by injecting 5 mL O_2/g VS, injected by once for 3 consecutive days. The overall increase compared to the untreated sample was 7.2%, but the difference was not statistically significant ($p > 0.05$). Moreover, the positive effect is markedly lower compared to a previous study, which found that equal micro-aeration intensity increased the biodegradability of corn straw by 16% (Fu et al., 2015). However, this difference in the micro-aeration efficiencies can be attributed to the chemical composition of corn straw, as it contains considerably lower lignin content (~11% TS) than wheat straw.

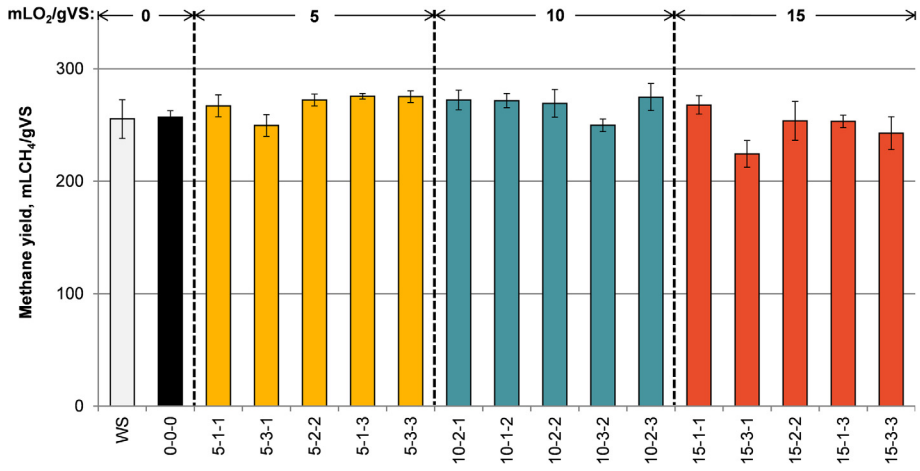


Fig. 3. Biochemical methane potential of wheat straw: untreated (i.e. WS and 0–0–0) and pretreated applying various micro-aeration strategies. The first bar depicts the AD of lignocellulosic biomass using only AD inoculum. The rest batch assays were conducted using a mixture of inocula. The treatments are denoted with the amount of oxygen load (mL O_2/g VS) – number of pulses – micro-aeration period (e.g. 5–1–3).

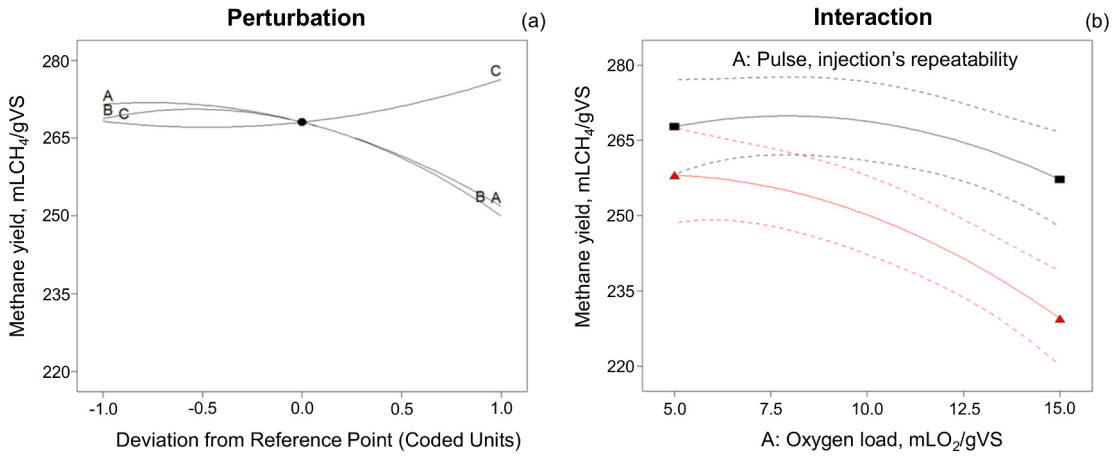


Fig. 4. Perturbation plot depicting the effect of process parameters on methane yield (a); interaction plot depicting the effect between oxygen load (A) and pulse repeatability (B) on methane yield (b).

Conversely, the highest adverse impact (−12.7%) was observed at the experimental set, in which the highest amount of oxygen was examined; supplied in 3 pulses for 1 treatment day ($p < 0.05$). The negative effect of micro-aeration can be attributed either to increased aerobic respiration or to the limited tolerance of the strict anaerobes to higher oxygen concentration (Fu et al., 2016).

The comparison of mean values indicated that the lowest load (5 mL O_2/g VS) was advantageous, resulting in the highest methane yield, while the highest oxygen load (15 mL O_2/g VS) had the opposite effect. Similarly to the present study, Mshandete et al. (2005) examined different aerobic conditions for enhancing the AD of sisal pulp waste under mesophilic conditions. They found that the productivity can be enhanced up to 26% under optimal conditions. However, a further 7-fold increase in the aeration level led to 37% lower methane production compared to the untreated samples and this was attributed to aerobic degradation. Thus, the findings from literature and the present study imply that the threshold level of micro-aeration intensity differs among different AD systems and should be always defined in order to avoid limited AD performance.

The second examined variable (i.e. pulse repeatability) showed a similar trend to the first variable (i.e. oxygen load), focusing on its different treatment levels. Specifically, injecting once or twice resulted in a similar positive level of methane enhancement ($p > 0.05$). Conversely, distributing the oxygen volume in 3 pulses had a slightly negative impact on methane yield (−3.4%) (Fig. 3).

Furthermore, it was demonstrated that the duration of the micro-aeration process was positively correlated with the methane production. More specifically, 3 days of treatment had the most positive impact on methane yield (264 ± 15 mL CH_4/g VS). Similarly, Lim and Wang (2013) found that the highest VFA accumulation was revealed on the third micro-aeration day out of a 4-days treatment period, which caused augmented substrate biodegradability. This result can be particularly associated with the finding of the present study (Fig. 2c), as the increased solubilization and VFAs concentration is pivotal in order to achieve enhanced methane production under micro-aerobic conditions.

3.3.2. Model estimation

The fit summary output of CCF revealed that the quadratic model was statistically significant for the methane yield, as the value of *F*-value (i.e. 16.89) was greater than the calculated one ($p < 0.0001$). The quadratic model had R^2 of 0.9383, adjusted-R^2 of 0.8827, predicted-R^2 of 0.6524 and adequate precision of 15.1449. The mentioned parameters indicated that the model can

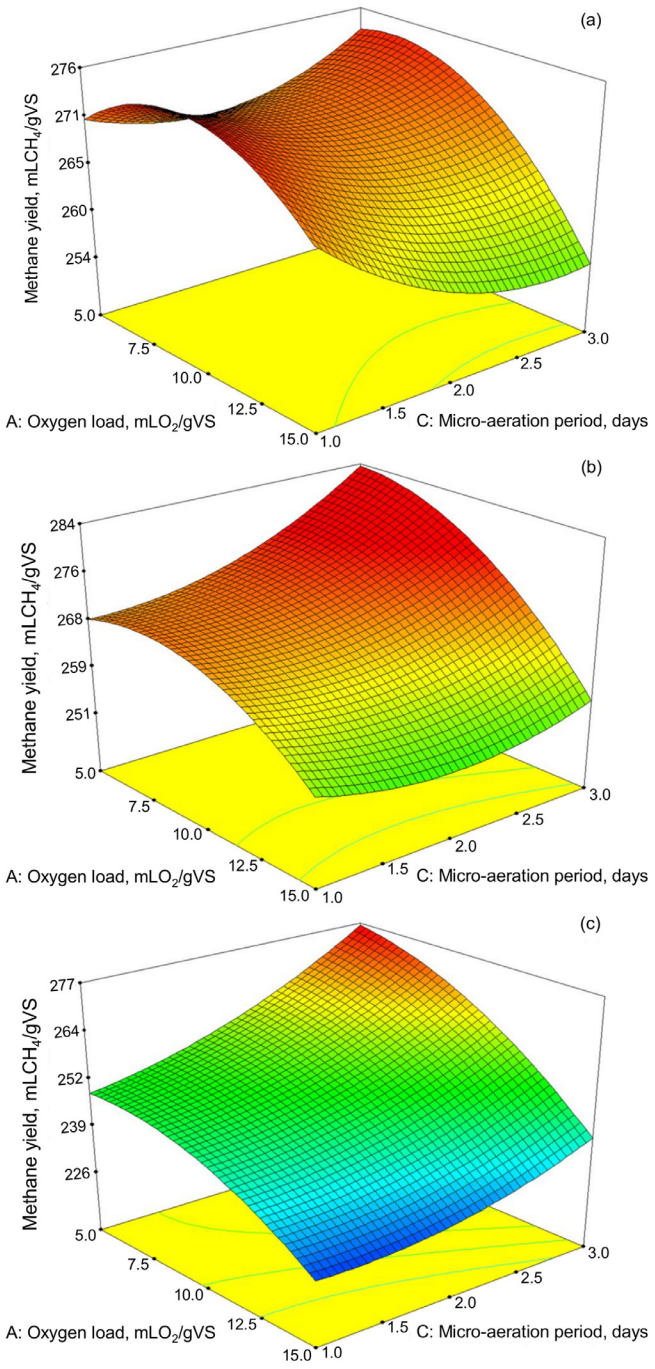


Fig. 5. Response graphs of methane yield as a function of two parameters by the quadratic model. The pulse repeatability was fixed at 1 injection (a), 2 injections (b) and 3 injections (c).

give a good fit prediction (Cornell, 2002). In general, the prediction quality is similar to the results of others researches, in which the optimisation of pretreatment methods was conducted using response surface methodology (Oliveira et al., 2014; Tedesco et al., 2014, 2013). Moreover, the results from the ANOVA revealed that apart from the second order of micro-aeration period (C^2), the rest examined variables significantly affected the methane yield. Nevertheless, the oxygen load (A) and pulse (B) were the most significant factors ($p < 0.0001$). However, it should be mentioned that the present quadratic model cannot be used for modelling future responses of alternative experimental set ups and it describes the results of the present BMP experiment. Overall, the final equations for methane yield in terms of coded (Eq. (1)) and actual (Eq. (2)) factors calculated by Design-Expert software are presented below.

$$CH_4\ yield = 268.054 - (9.837 * A) - (9.395 * B) + (4.035 * C) - (4.542 * A * B) - (3.834 * AC) + (6.300 * BC) - (6.403 * A^2) - (8.727 * B^2) + (4.221 * C^2) \quad (1)$$

$$CH_4\ yield = 246.506 + (6.505 * \text{Oxygen load}) + (21.998 * \text{Pulse}) - (17.779 * \text{Micro-aeration period}) - (0.908 * \text{Oxygen load} * \text{Pulse}) - (0.767 * \text{Oxygen load} * \text{Micro-aeration period}) + (6.300 * \text{Pulse} * \text{Micro-aeration period}) - (0.256 * \text{Oxygen load}^2) - (8.727 * \text{Pulse}^2) + (4.221 * \text{Micro-aeration period}^2) \quad (2)$$

The perturbation plot in Fig. 4a shows that factors A (oxygen load) and B (pulse) influence differently the methane yield compared to C (micro-aeration period). Oxygen load and pulse reduction from the reference point can have a positive impact on methane yield and conversely, their increase affect negatively the methane response. On the other hand, the micro-aeration period had a convex effect on the methane yield and thus, through extending the micro-aeration period, a positive effect on the methane response could be achieved. The plot in Fig. 4b demonstrates that a marked interaction could occur between A and B in the region of an oxygen load between 5 and 7.5 mL O_2/g VS defined by the interacted confidence bands. Also, it is observed that oxygen loads higher than 10 mL O_2/g VS can potentially result in a significant decrease in the methane yield (Fig. 4b).

3.3.3. Optimisation of methane yield

The optimisation case study was performed with numerical limiting criteria. The effects of oxygen load and micro-aeration period on methane yield at the 3 examined pulse injections' repetition are graphically represented in Fig. 5.

It can be observed that a consecutive 3-day treatment period affects positively the methane production (i.e. augmented red area) (Fig. 5). Additionally, as explained above, the optimum values for the physicochemical characteristics (i.e. increased sCOD and VFA) were obtained on the third day (Fig. 2b and c). The response graphs differ visually, clearly indicating that the distribution of oxygen volume in three injections (Fig. 5c) could lower the methane yield, as the target area (red color) covers limited surface. In contrast, the highest biodegradability levels are predicted by injecting 2 pulses of oxygen load (Fig. 5b).

Therefore, the recommended micro-aeration strategy includes the injection of 7.3 mL O_2/g VS distributed in 2 pulses for a period that will last slightly less than 3 days (i.e. 47 h). The recommended strategy could lead to a more distinct impact compared with the experimentally tested values, as the anaerobic biodegradability was calculated to be increased by 9% compared to the untreated wheat straw (281 mL CH_4/g VS). Likewise, the same level of enhancement was observed when brown water and food waste were micro-aerated by injecting 37.5 mLO_2/L/day (Lim and Wang, 2013). Interestingly, this oxygen volume is comparable to the suggested strategy of the optimisation study (i.e. 7.3 mL O_2/g VS or 36.5 mLO_2/L/day). Hence, BMP results and the optimisation study imply that oxygen poses the ability to be considered as an inexpensive and easily applicable method in order to efficiently improve the anaerobic biodegradability of recalcitrant biomasses.

4. Conclusions

A number of micro-aeration strategies using a mixture of aerobic and anaerobic inocula revealed different results on the biomethanation process of wheat straw. The highest enhancement of methane yield (+7.2%) was achieved by injecting pulses of 5 mL O_2/g VS, for a 3-day period. Higher oxygen loads could lead to negative impact on the biomethanation process. Finally, the optimisation study concluded that the methane yield could be further improved following an alternative suggested micro-aeration strategy.

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